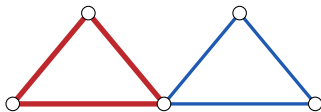


Trails, Circuits, Paths, and Cycles

Section 1.4

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



Jeremy Quastel

Today's lecture

1. The Bridges of Königsberg
Section 1.4.1.

2. Eulerian Trails and Circuits
Section 1.4.2.

**3. Hamiltonian Paths
and Cycles**
Section 1.4.3.

4. Three Open Problems
Section 1.4.4.

Two Route-Planning Questions

Question 1: Covering streets

Can a route use every street exactly once?

Question 2: Visiting towns

Can a route visit every town exactly once?

The first question concerns **edges**; the second concerns **vertices**.

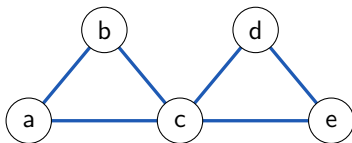
How different can the answers be?

Recall: Walks, Trails, Paths, and Cycles

Walk	Consecutive vertices are adjacent.
Trail	No edge is repeated.
Circuit	A closed trail.
Path	No vertex is repeated.
Cycle	A closed path, except for the repeated endpoint.

A circuit may revisit a vertex. A cycle cannot revisit a vertex other than its starting vertex at the end.

A Circuit Need Not Be a Cycle



Example

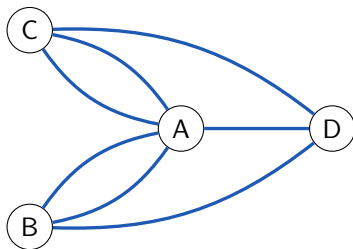
a, b, c, d, e, c, a is a circuit.

It is not a cycle: the vertex c is visited twice before returning to a .

The Bridges of Königsberg

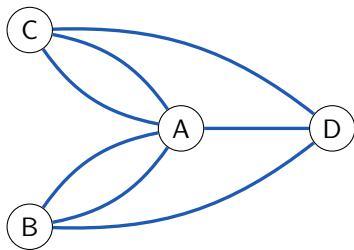
Question

Can we cross each of the seven bridges exactly once?



Each vertex represents a land mass; each edge represents a bridge.
The route may start and finish at different vertices.

Representing a Route by Letters



The route A, B, D, A, C, A, B uses six bridges.

A route crossing all seven bridges would use **eight letters**.

Parallel edges represent different bridges, even when their endpoints agree.

Euler's Counting Idea

How many times must each letter appear?

A vertex with $2k + 1$ incident bridges must be an endpoint of the route and must appear $k + 1$ times in its letter sequence.

Here the degrees are 5, 3, 3, 3. Thus the required occurrences are

$$3 + 2 + 2 + 2 = 9.$$

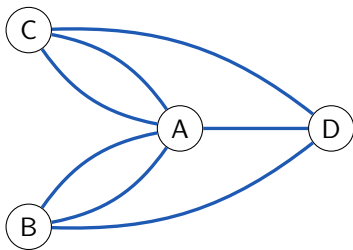
But a seven-bridge route has only eight letters.

No such route exists.

The Same Obstruction: Odd Degrees

Entering and leaving

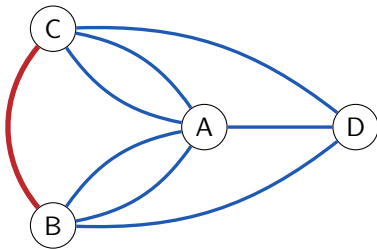
At every vertex other than the endpoints, edges used to enter and leave come in pairs.



There can be at most two vertices of odd degree.

Königsberg has four.

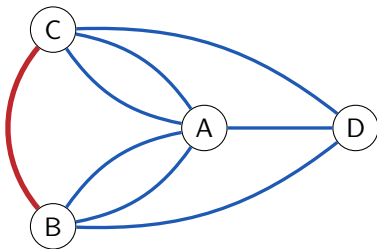
What If We Add One Bridge?



Question

Add a bridge between B and C . Can we now use every bridge exactly once? Where must the route start and end?

What If We Add One Bridge? Answer



Only A and D now have odd degree.

One route is

$A, B, A, C, A, D, B, C, D.$

The two AB edges and the two AC edges are distinct bridges.

Definition: Eulerian Trails and Circuits

Eulerian trail

A trail using every edge of the graph.

Eulerian circuit

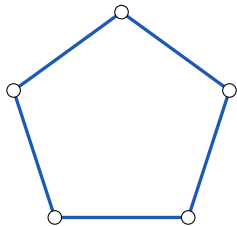
A circuit using every edge of the graph.

Eulerian graph

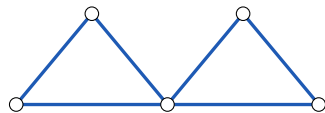
A graph containing an Eulerian circuit.

Vertices may repeat. Edges may not.

Examples: Which Graphs Are Eulerian?



C_5

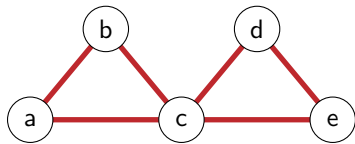
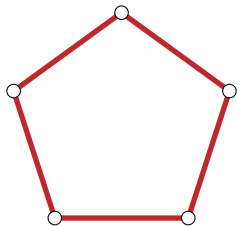


Two triangles sharing a vertex

Question

Does either graph have an Eulerian circuit? Can you describe a route?

Examples: Eulerian Circuits



Both graphs have Eulerian circuits.

In the right-hand graph, use *a*, *b*, *c*, *d*, *e*, *c*, *a*.

A repeated vertex is allowed in an Eulerian circuit.

Theorem: Characterizing Eulerian Graphs

Theorem 1.20

For a connected graph G , the following are equivalent.

- ① G has an Eulerian circuit.
- ② Every vertex has even degree.
- ③ $E(G)$ is a disjoint union of edge sets of cycles.

Proof plan: $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$.

Proof of Theorem 1.20: Even Degrees

Recall

An Eulerian circuit forces all degrees to be even.

Proof

Follow the circuit from its starting vertex. Each arrival is paired with a departure, using two distinct edges. At the starting vertex, pair the first departure with the final arrival.

Every incident edge is used exactly once, so the degree of every vertex is even.



Proof Continued: Partitioning into Cycles

Recall

If all degrees are even, the edges can be partitioned into cycles.

Proof

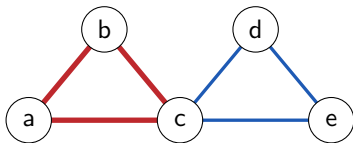
If edges remain, some nontrivial component has minimum degree at least 2, so it contains a cycle S . Remove the edges of S .

Each vertex of S loses two incident edges; every degree remains even. Repeat on the remaining graph.

The number of edges decreases, so the procedure ends. The removed cycles partition the original edge set. □

Isolated vertices contribute no edges; the empty edge set has the empty partition.

Partitioning into Cycles: Example



The red triangle and the blue triangle have disjoint edge sets.

They share the vertex c , which is allowed.

Removing the red cycle changes the degree of c from 4 to 2.

Proof Continued: Joining the Cycles

Recall

An edge partition into cycles gives an Eulerian circuit.

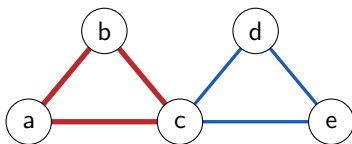
Proof

Choose a circuit C using the edges of as many whole cycles of the partition as possible.

If an edge remains unused, connectedness gives an unused edge incident with a vertex v of C . The partition cycle S containing that edge is entirely unused.

At v , traverse S , return to v , and then continue around C . This is a larger circuit, a contradiction. □

Joining Two Circuits



Start with a, b, c, a . Insert c, d, e, c at c :

$$a, b, \underbrace{c, d, e, c}_{\text{inserted circuit}}, a.$$

No edge is repeated, because the two circuits have disjoint edge sets.

Corollary: Eulerian Trails

Corollary 1.21

A connected graph has an Eulerian trail exactly when it has at most two vertices of odd degree.

- No odd-degree vertices: the route can be closed.
- Exactly two odd-degree vertices: they are the endpoints.
- More than two odd-degree vertices: no Eulerian trail.

Recall: the number of odd-degree vertices is always even.

Proof of Corollary 1.21

Recall

A connected graph has an Eulerian trail iff it has 0 or 2 odd-degree vertices.

Proof

Necessity. Pair arrivals and departures at all internal vertices. Only the two endpoints can have odd degree.

Sufficiency. With no odd degrees, use Theorem 1.20. With odd vertices u, v , add a new vertex x and edges ux, xv . All degrees are now even.

Take an Eulerian circuit and remove the segment u, x, v . The remaining route is an Eulerian trail from v to u . □

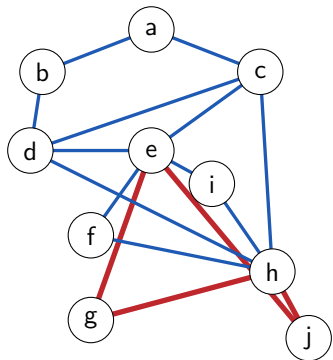
Hierholzer's Algorithm

Once we know an Eulerian circuit exists, how do we find one?

Algorithm

- 1 Build a circuit and mark its edges.
- 2 If all edges are marked, stop.
- 3 Choose a vertex on the circuit incident with an unmarked edge.
- 4 From that vertex, build a circuit using unmarked edges.
- 5 Insert the new circuit into the old one and repeat.

Hierholzer's Algorithm: Step 1



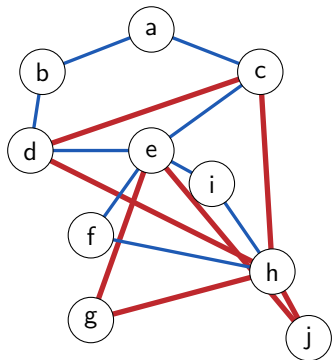
Choose $R_1 = e, g, h, j, e$.

Current circuit

$R_1 = e, g, h, j, e$.

Red edges have been used. Blue edges remain.

Hierholzer's Algorithm: Step 2



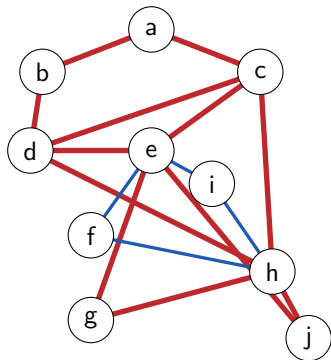
At h , insert $Q_1 = h, d, c, h$.

Current circuit

$R_2 = e, g, h, d, c, h, j, e$.

Red edges have been used. Blue edges remain.

Hierholzer's Algorithm: Step 3



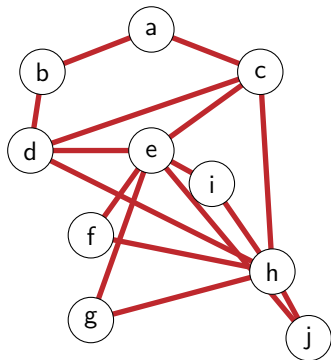
At d , insert $Q_2 = d, b, a, c, e, d$.

Current circuit

$R_3 = e, g, h, d, b, a, c, e, d, c, h, j, e$.

Red edges have been used. Blue edges remain.

Hierholzer's Algorithm: Step 4



At h , insert $Q_3 = h, f, e, i, h$.

Current circuit

$R_4 = e, g, h, f, e, i, h, d, b, a, c, e, d, c, h, j, e$.

All 16 edges are now used.

Why Hierholzer's Algorithm Works

- Removing a circuit leaves every remaining degree even.
- A trail following unused edges can get stuck only at its starting vertex; this produces another circuit.
- If unused edges remain, connectedness gives a place to attach another circuit.
- Each insertion adds edges. Since the graph is finite, the algorithm stops.

From Edges to Vertices

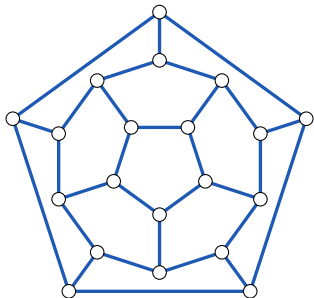
A different question

Can we visit every vertex exactly once, and return to the start?

Hamilton's Icosian game asks for such a route on the graph of a dodecahedron.

We now seek a **spanning cycle**, rather than a circuit using every edge.

The Icosian Game

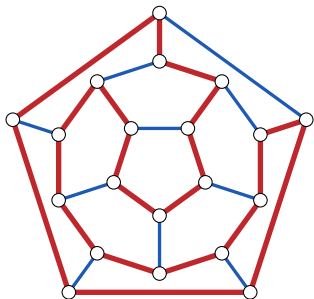


Question

Find a cycle through all twenty vertices of the dodecahedral graph.

The drawing has been flattened into the plane. Adjacencies remain the same.

The Icosian Game: A Solution



The red cycle visits each vertex exactly once before returning to the start.

It uses 20 of the graph's 30 edges.

This is the type of route we call Hamiltonian.

Definition: Hamiltonian Paths and Cycles

Hamiltonian path and traceable graph

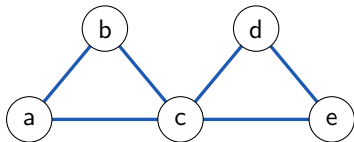
A Hamiltonian path contains every vertex. A graph with such a path is called traceable.

Hamiltonian cycle and Hamiltonian graph

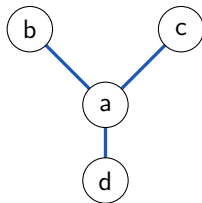
A Hamiltonian cycle contains every vertex. A graph with such a cycle is called Hamiltonian.

Every Hamiltonian graph is traceable: remove one edge of its spanning cycle.

Examples: Hamiltonian, Traceable, or Neither?



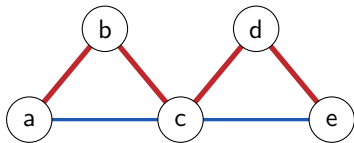
Two triangles sharing a vertex



Question

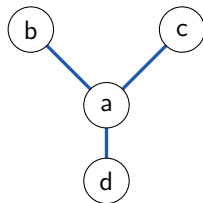
Which graphs have spanning paths? Which have spanning cycles?

Examples: Answers



Left: traceable via a, b, c, d, e , but not Hamiltonian. A cycle cannot visit both triangles without repeating c .

Right: neither. A spanning path would need all three leaves as endpoints.



Eulerian and Hamiltonian Are Different

Graph	Eulerian?	Hamiltonian?
C_5	Yes	Yes
Two triangles sharing a vertex	Yes	No
K_4	No	Yes
P_4	No	No

Degree parity completely determines Eulerian circuits in connected graphs. It does not determine Hamiltonian cycles.

Hamiltonian Graphs Must Be 2-Connected

Observation

If G is Hamiltonian, then deleting any one vertex leaves a connected graph.

Delete a vertex from a Hamiltonian cycle. The remaining edges of the cycle form a path through all remaining vertices.

Thus a cut vertex rules out Hamiltonicity.

The reverse implication is false.

Theorem: Dirac's Theorem

Theorem 1.22

For a graph of order $n \geq 3$, $\delta(G) \geq n/2$ implies that G is Hamiltonian.

Large degrees can force a spanning cycle.

We prove this by turning a longest path into a cycle.

Proof of Dirac's Theorem: A Longest Path

Recall

For a graph of order $n \geq 3$, $\delta(G) \geq n/2$ implies that G is Hamiltonian.

Proof

First, G is connected: each component has at least $\delta(G) + 1 > n/2$ vertices, so two components are impossible.

Choose a longest path $P = v_1, v_2, \dots, v_p$. Every neighbor of either endpoint lies on P ; otherwise the path could be extended.

Proof Continued: Finding Two Cross Edges

Recall

$P = v_1, \dots, v_p$ is a longest path.

Proof

In $\{1, \dots, p-1\}$, define

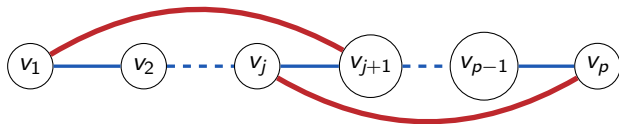
$$A = \{i : v_1 v_{i+1} \in E(G)\}, \quad B = \{i : v_i v_p \in E(G)\}.$$

All endpoint neighbors lie on P , so

$$|A| + |B| = \deg(v_1) + \deg(v_p) \geq n > p - 1.$$

Thus $A \cap B \neq \emptyset$. For some j , both $v_1 v_{j+1}$ and $v_j v_p$ are edges.

Proof Continued: Closing the Path



Follow the vertices in this order:

$$v_1, v_2, \dots, v_j, v_p, v_{p-1}, \dots, v_{j+1}, v_1.$$

This is a cycle containing every vertex of P .

Proof Continued: The Cycle Is Spanning

Recall

There is a cycle C containing every vertex of a longest path P .

Proof

If C misses a vertex, connectedness gives an edge xy with $x \notin V(C)$ and $y \in V(C)$.

Start at x , move to y , and follow C without closing it. This gives a path with $p + 1$ vertices, contradicting the choice of P .

Hence $p = n$, and C is Hamiltonian. □

Dirac's Bound: Sharp, but Not Necessary

Sharpness

$K_{r,r+1}$ has $n = 2r + 1$ and minimum degree $r = (n - 1)/2$. It has no Hamiltonian cycle: a cycle in a bipartite graph uses equal numbers of vertices from the two parts.

Not a characterization

For $n \geq 5$, the graph C_n is Hamiltonian but has minimum degree $2 < n/2$.

Theorem: Ore's Theorem

Theorem 1.23

Let $|V(G)| = n \geq 3$. If

$$\deg(x) + \deg(y) \geq n$$

for every pair of nonadjacent vertices x, y , then G is Hamiltonian.

Dirac's condition implies this condition.

Exercise: adapt the longest-path argument to prove it.

Definitions: Independence and Connectivity

Independence number

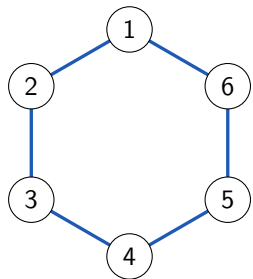
A set of vertices is independent if no two of its vertices are adjacent.

$$\alpha(G) = \max\{|S| : S \subseteq V(G) \text{ is independent}\}.$$

Vertex connectivity

$\kappa(G)$ is the minimum number of vertices whose removal disconnects G or leaves one vertex. In particular, $\kappa(K_n) = n - 1$.

Example: Independence Number



In C_6 , the sets $\{1, 3, 5\}$ and $\{2, 4, 6\}$ are independent.

No independent set can use more than one endpoint of each of the disjoint edges 12, 34, 56.

Thus $\alpha(C_6) = 3$.

Also, $\kappa(C_6) = 2$.

Theorem: Connectivity and Independence

Theorem 1.24 (Chvátal–Erdős)

For a connected graph of order at least 3, $\kappa(G) \geq \alpha(G)$ implies Hamiltonicity.

High connectivity prevents a longest cycle from missing a component unless a large independent set exists.

Proof of Theorem 1.24: Set-Up

Recall

For a connected graph of order at least 3, $\kappa(G) \geq \alpha(G)$ implies Hamiltonicity.

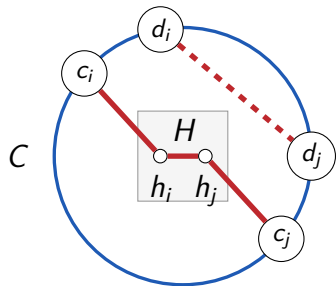
Proof

The hypothesis implies $\kappa(G) \geq 2$: otherwise $\alpha(G) \leq 1$, so G is complete, contradicting $\kappa(G) = 1$ and $n \geq 3$.

Let C be a longest cycle. If it is not spanning, choose a component H of $G - V(C)$.

Let $c_1, \dots, c_r$ be the vertices on C with neighbors in H , in cyclic order. Let d_i be the successor of c_i on the oriented cycle.

Proof Continued: Successors of Attachment Vertices

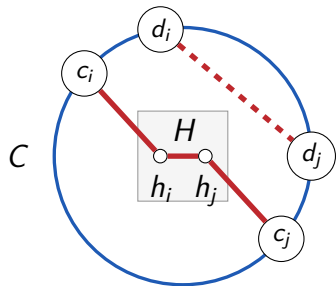


Choose $h_i \in H$ adjacent to c_i .

Observation 1. No d_i has a neighbor in H . Otherwise replace $c_i d_i$ by a path through H to obtain a longer cycle.

Observation 2. Removing $c_1, \dots, c_r$ separates H from the successors on C . Hence $r \geq \kappa(G)$.

Proof Continued: The Successors Are Independent



Observation 3. No two successors d_i, d_j are adjacent.

If they were, use a path through H from c_i to c_j , then follow C backwards to d_i , cross to d_j , and follow C forwards to c_i .

This cycle contains all of C and at least one vertex of H , a contradiction.

Proof Continued: The Contradiction

Recall

For a connected graph of order at least 3, $\kappa(G) \geq \alpha(G)$ implies Hamiltonicity.

Proof

Choose any vertex $h \in H$. The observations show that

$$\{h, d_1, \dots, d_r\}$$

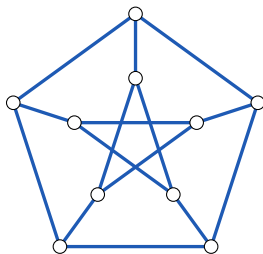
is independent: no two successors are adjacent, and no successor has a neighbor in H .

Consequently,

$$\alpha(G) \geq r + 1 \geq \kappa(G) + 1,$$

contrary to $\kappa(G) \geq \alpha(G)$. Thus the longest cycle is spanning. □

The Petersen Graph



The Petersen graph has

$$\kappa(G) = 3, \quad \alpha(G) = 4.$$

It is not Hamiltonian.

Thus replacing $\kappa(G) \geq \alpha(G)$ by $\kappa(G) \geq \alpha(G) - 1$ would not work.

Exercise: find an independent set of size 4.

Definition: Forbidden Induced Subgraphs

H -free

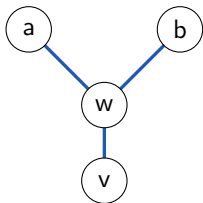
A graph G is H -free if no induced subgraph of G is isomorphic to H .

$\mathcal{S}$ -free

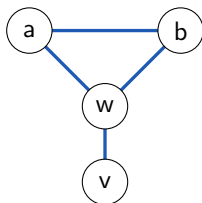
A graph is $\mathcal{S}$ -free if it is H -free for every $H \in \mathcal{S}$.

Induced matters: select vertices and retain every edge between them. Extra edges can prevent an induced copy.

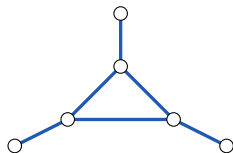
Three Important Forbidden Graphs



Claw: $K_{1,3}$



Z_1 : a triangle with a leaf

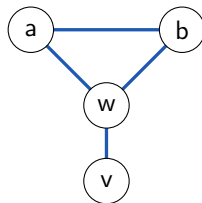
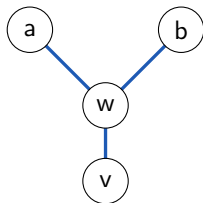


N : a triangle with three leaves

Theorem: Two Forbidden Subgraphs

Theorem 1.25

Every 2-connected $\{K_{1,3}, Z_1\}$ -free graph is Hamiltonian.



Proof of Theorem 1.25: A Longest Cycle

Recall

Every 2-connected $\{K_{1,3}, Z_1\}$ -free graph is Hamiltonian.

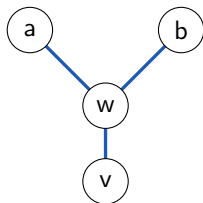
Proof

Let C be a longest cycle. If C is not spanning, connectedness gives a vertex $v \notin C$ adjacent to $w \in C$.

Let a, b be the predecessor and successor of w on C . Neither av nor bv can be an edge: either would let us insert v into C .

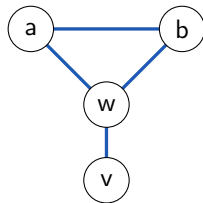
We now consider whether a and b are adjacent.

Proof Continued: Two Cases



Case 1: ab is absent

The vertices w, a, b, v induce a claw.



Case 2: ab is present

The vertices w, a, b, v induce Z_1 .

Both cases contradict the forbidden-subgraph hypothesis.

Hence C is Hamiltonian.

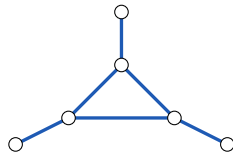
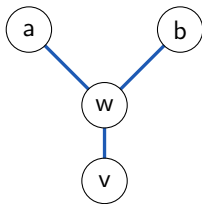


Theorem: Forbidding the Claw and the Net

Theorem 1.26

For a $\{K_{1,3}, N\}$ -free graph G :

- 1 Connectedness implies traceability.
- 2 2-connectedness implies Hamiltonicity.



Three Open Problems

The final part of the section moves from theorems to conjectures.

- Do any three longest paths share a vertex?
- Does 4-connectivity force a claw-free graph to be Hamiltonian?
- Can the vertex set always be partitioned with prescribed bounds on longest paths?

We follow the problems and numbering of Section 1.4.4.

Definition: Detour Order and Detour Path

Detour order

$\tau(G)$ is the number of vertices in a longest path of G .

Detour path

A path with $\tau(G)$ vertices is called a detour path.

For example, $\tau(P_n) = n$, $\tau(C_n) = n$, and $\tau(K_{1,3}) = 3$.

Detour order counts **vertices**, rather than edges.

Any Two Longest Paths Intersect

Claim

Any two longest paths in a connected graph share a vertex.

Proof

Suppose disjoint longest paths P, Q each have L edges. Choose a shortest path joining them, with endpoints $x \in P$ and $y \in Q$ and no internal vertex on either.

From x , one half of P has at least $L/2$ edges; similarly for Q at y . Joining these two halves gives a path of length at least $L + 1$, a contradiction. □

What About Three Longest Paths?

Conjecture 1

For three detour paths P_1, P_2, P_3 in a connected graph,

$$V(P_1) \cap V(P_2) \cap V(P_3) \neq \emptyset.$$

Pairwise intersection alone does not imply a common vertex.

The question asks whether being *longest paths in the same graph* supplies the missing structure.

All Longest Paths Need Not Share a Vertex

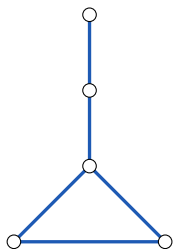
The analogous statement for the entire family of longest paths is false.

Examples discussed in the textbook

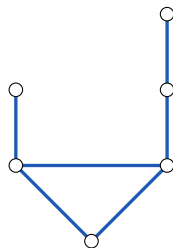
The section gives a 12-vertex graph with detour order 10, and another graph with seven detour paths whose common intersection is empty.

These examples explain why the question about **three** paths is more delicate than the question about all paths.

Two More Forbidden Graphs



Z_2



W

Each starts with a triangle. In Z_2 , attach a two-edge path at one vertex. In W , also attach a leaf at another triangle vertex.

Why Does the Claw Keep Appearing?

Theorem 1.27

Let R, S be connected graphs other than P_3 . If forbidding R, S as induced subgraphs forces every 2-connected graph to be Hamiltonian, then one of R, S is $K_{1,3}$ and the other is an induced subgraph of P_6, Z_2, W , or N .

The graphs Z_2 and W are shown on the previous slide.

The claw is an essential part of such forbidden pairs.

Matthews–Sumner's Conjecture

Conjecture 2

Every 4-connected claw-free graph is Hamiltonian.

3-connectivity is insufficient: there are 3-connected claw-free non-Hamiltonian graphs.

Theorem 1.28 (as stated in the textbook)

Every 7-connected claw-free graph is Hamiltonian.

Later work improves the sufficient connectivity from 7 to 6.

See Allred and Ellingham, *Unavoidable induced subgraphs of large and infinite 2-edge-connected graphs*, 2026, Section 1.2.

Definition: An (a, b) -Partition

Write $\langle A \rangle$ for the subgraph induced by a vertex set A .

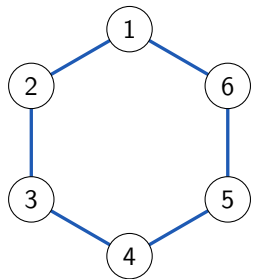
(a, b) -partition

A partition $V(G) = A \sqcup B$ such that

$$\tau(\langle A \rangle) \leq a, \quad \tau(\langle B \rangle) \leq b.$$

The bounds concern longest paths within the two induced graphs. There may be many edges between A and B .

Example: Partitioning a Cycle



Here $\tau(C_6) = 6$.

Let $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$. Both induced graphs have no edges, so both have detour order 1.

This is a valid (a, b) -partition for every positive a, b with $a + b = 6$.

The Path Partition Conjecture

τ -partitionable

G is τ -partitionable if it has an (a, b) -partition for every pair of positive integers satisfying $a + b = \tau(G)$.

Path Partition Conjecture

Every graph is τ -partitionable.

The partition may depend on (a, b) .

One fixed partition need not work for every pair.

Known Cases of the Path Partition Conjecture

Theorem 1.29

G is τ -partitionable if any of these conditions holds:

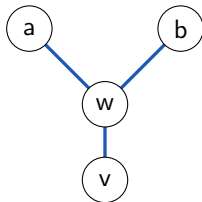
- 1 $\Delta(G) \leq 3$;
- 2 $\Delta(G) \geq |V(G)| - 8$;
- 3 $\tau(G) \leq 13$;
- 4 $\tau(G) \geq |V(G)| - 1$.

These are the sufficient conditions listed in the textbook.

The Claw Appears Again

Theorem 1.30

Every claw-free graph is τ -partitionable.



The same small forbidden graph appears in both Hamiltonicity and path-partition results.

Summary: Trails, Circuits, Paths, and Cycles

- **Eulerian routes:** use every edge; degree parity gives a complete criterion.
- **Finding circuits:** Hierholzer's algorithm joins circuits along shared vertices.
- **Hamiltonian routes:** visit every vertex; degree bounds, connectivity, and forbidden induced subgraphs give sufficient conditions.
- **Further questions:** intersecting longest paths, claw-free Hamiltonicity, and path partitions.

Planarity